

# ZACHARY DUTTON

Additive Manufacturing | Mechanical Design | Engineering Support

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## WORK & INTERNSHIP EXPERIENCE

### Sig Sauer - Additive Manufacturing Intern (3 terms) | Exeter, NH

May 2024 - Aug 2025

*Returning intern: May - Aug 2024; Dec 2024 - Jan 2025 (winter break); May - Aug 2025*

- Improved manufacturability across the metal AM cell through time and powder studies on reclaim, build plate storage and removal, M290 recoater blade selection, and textured surface removal; built quality measurement matrices to track laser run-off.
- Operated EOS M290/M400 DMLS systems as the cell grew from three machines to seven; helped bring three new M400s online with Grenzebach pallet exchangers for automatic build changes and continuous operation.
- Helped integrate a Volkmann enclosed depowdering and powder reclaim system; improved the wire EDM, ultrasonic, and manual cleaning workflow to reduce powder residue and updated control plans and work instructions.
- Helped get a used, non-running EOS M290 working in a fast-paced R&D environment; resurfaced corroded coolant sealing faces in-house instead of waiting months for the OEM repair, with no leaks afterward.
- Designed build plates and fixtures, and redesigned a worn M290 latch for a tighter closure, printed in 17-4 stainless steel, so the machine passed its pressure and purge tests.

### Pratt & Whitney - Aftermarket Sustainment Engineer Co-op | North Berwick, ME

Jan 2025 - May 2025

- Worked One-Time Concessions and Long-Term Repairs for hot-section hardware on 24K/30K commercial engines, mainly split cases and stator rings, working alongside the disciplines each case needed to verify whether damage could pass as-is or be repaired.
- Partnered with repair shops worldwide to identify recurring issues and develop repairs such as plasma spray, welding, and blending; drafted repair instructions and supporting documentation through ODA sign-off.

### Pirouette Pharma Inc. - Engineering Intern / Assembly Technician | Portsmouth, NH

May 2023 - Aug 2023

- Helped construct a purchased cleanroom and set up an APSX benchtop injection molder to run Formlabs 3D-printed resin molds.
- Ran pressure decay, occlusion, and functional testing on medical device test articles, plus heat-accelerated aging, Instron tensile tests, and process time trials; contacted vendors for ASTM vibration testing.

### Dover Honda - Express Lube Technician | Dover, NH

Jun 2022 - Aug 2022

## SELECTED PROJECTS

### Sig Sauer Coolant Stand - Structural Analysis

June 2025

- Ran linear static FEA in SolidWorks on a colleague's welded steel stand under 10,000 lbf; predicted about 4 mm deflection.
- Traced the model's 0.507 minimum safety factor to a single node at a bonded contact on an anti-crush sleeve; the structure itself held a safety factor above 10, and the stand passed EH&S review.

### Scan-to-CAD Vacuum Accessory Dock

Personal project

- Rebuilt a 3D-scanned accessory interface as editable CAD (RevoScan, Solid Edge, Fusion), then designed and printed the dock.

## TECHNICAL SKILLS

|                        |                                                                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAD & Analysis         | SolidWorks, SolidWorks Simulation (static FEA), Siemens NX, CATIA, Fusion, Inventor, Solid Edge; technical drawings, GD&T; Revopoint 3D scanning and scan-to-CAD reverse engineering                                        |
| Additive Manufacturing | EOS M290/M400, DMLS/LPBF, build preparation, powder handling, post-processing, Grenzebach pallet automation, Volkmann depowdering and powder reclaim, Cura slicer; resin, SLS, FDM                                          |
| Manufacturing          | Tooling/fixtures, wire EDM, basic CNC toolpaths and manual G-code, milling/lathe, work instructions, 6S                                                                                                                     |
| Testing & Metrology    | Pressure decay, occlusion, tensile/compressive (Instron UTM), heat-accelerated aging, and Rockwell hardness testing; calipers, bore calipers, digital and height micrometers, gauge pins, feeler gauges, optical micrometer |
| Software               | Microsoft Office (advanced Excel with VBA), Python                                                                                                                                                                          |

## EDUCATION

University of Central Florida, Orlando, FL - Engineering coursework

Embry-Riddle Aeronautical University, Daytona Beach, FL - Engineering coursework

Completing bachelor's degree part time through online coursework.

UCF Baja Club: SolidWorks body wireframe CAD support.

## CERTIFICATIONS & TRAINING

SolidWorks CSWA | OSHA 10-Hour Safety Course

GD&T training per ASME Y14.5-2018 (GeoTol Pro 2020) - Completed during Pratt & Whitney co-op.